IMPERIAL

On the connection between the action of $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$ and the classification of lattices $\Lambda \subset \mathbb{C}$

The action of $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$

Definition 1. Let $\mathbb{H}$ denote the upper half of the complex plane:

$$\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$$

Let $\mathrm{SL}(2,\mathbb{Z})$ denote the group of matrices of $\mathbb{Z}^{2\times 2}$ with determinant 1. We define $\mathrm{PSL}(2,\mathbb{Z}) = \mathrm{SL}(2,\mathbb{Z})/\{\pm I\}$. For each $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}(2,\mathbb{Z})$, it acts on $\mathbb{H}$ via the corresponding Möbius map:

$$z \mapsto \frac{az+b}{cz+d}$$

Theorem 2. $\mathrm{PSL}(2,\mathbb{Z})$ is generated by the Möbius maps $z \mapsto z+1$ and $z \mapsto -1/z$.

Proof. Let

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

be the matrices corresponding to $z \mapsto z+1$ and $z \mapsto -1/z$, respectively.

For any matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}(2,\mathbb{Z})$. If $|a| \geq |c| > 0$, there exists unique $k, r \in \mathbb{Z}$ such that $a = kc + r$ where $0 \leq r < |c| \leq |a|$. It follows that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = T^k \begin{pmatrix} r & b-kd \\ c & d \end{pmatrix}$$

If $|c| > |a| > 0$, there exists unique $k, r \in \mathbb{Z}$ such that $c = ka + r$ where $0 \leq r < |a|$. It follows that

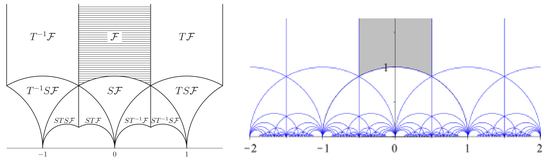
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = S \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = ST^{-k} \begin{pmatrix} r & d-kb \\ -a & -b \end{pmatrix}$$

Repeatedly apply the two steps above, which will terminate in a finite number of times since $|a|$ decreases each time. M then equals a combination of T, S and a matrix with $a = 0$ or $c = 0$, which is obviously generated by T and S. We conclude that $M \in \langle T, S\rangle$ and then $\langle T, S\rangle = \mathrm{PSL}(2,\mathbb{Z})$. □

Definition 3. By a *fundamental domain* for $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ we mean an open subset $\mathcal{F} \subset \mathbb{H}$ where $\mathcal{F} \cap \gamma\mathcal{F} = \varnothing$ for $\gamma \in \mathrm{PSL}(2,\mathbb{Z}) \setminus \{I\}$ and $\bigcup \gamma\overline{F} = \mathbb{H}$.[1]

Figures below visualize the action of $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$ and the standard fundamental domain $\mathcal{F}$ (shaded area) given by

$$\mathcal{F} = \left\{ z \in H : |\Re z| < \frac{1}{2}, |z| > 1 \right\}$$



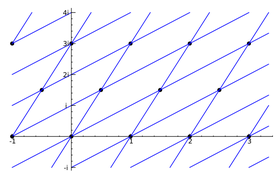
Each hyperbolic triangle in the images can be translated from the fundamental domain via a Möbius map in $\mathrm{PSL}(2,\mathbb{Z})$.

Lattices $\Lambda \subset \mathbb{C}$

Definition 4. A *lattice* Λ in the complex plane is an additive discrete subgroup of $\mathbb{C}$. [2] Equivalently, $\Lambda \subset \mathbb{C}$ is a lattice if there are $p, q \in \mathbb{C}$ linearly independent over $\mathbb{R}$, i.e. $\frac{p}{q} \notin \mathbb{R}$, such that

$$\Lambda = \langle p, q\rangle = \{ap + bq : a, b \in \mathbb{Z}\}$$

As an example, the figure below shows the lattice $\langle \frac{1}{2} + \frac{3}{2}i, \frac{3}{2} + \frac{3}{2}i\rangle$.



For a lattice $\Lambda = \langle p, q\rangle$, one can prove that the parallelogram

$$\mathcal{F} = \{ap + bq : 0 \leq a, b < 1\}$$

is a fundamental domain for $\mathbb{C}/\Lambda$.

Theorem 5. [2] Let $\Lambda = \langle p, q\rangle$ and $\Lambda' = \langle p', q'\rangle$ be two lattices with $\frac{p}{q}, \frac{p'}{q'} \in \mathbb{H}$. We have $\Lambda = \Lambda'$ if and only if there is a matrix $M \in \mathrm{SL}(2,\mathbb{Z})$ such that

$$\begin{pmatrix} p' \\ q' \end{pmatrix} = M \begin{pmatrix} p \\ q \end{pmatrix}$$

Homothetic lattices

We want to regard two lattices to be 'similar' if one can be converted to the other via some basic geometric transformations which preserves most of the properties of the lattice structure. Under this idea, we give the following definition.

Definition 6. Two lattices Λ and Λ' are said to be *homothetic* if there is $c \in \mathbb{C} \setminus \{0\}$ such that $\Lambda' = c\Lambda$. [1]

It's easy to see that, if $\Lambda' = re^{i\theta}\Lambda$ are homothetic, then Λ can be transformed to Λ' via a dilation by r and a rotation by θ.

For the structure-preserving property, one has the following theorem.

Theorem 7. Let Λ and Λ' be two lattices in $\mathbb{C}$. Then $\mathbb{C}/\Lambda$ is isomorphic to $\mathbb{C}/\Lambda'$ if and only if Λ and Λ' are homothetic. [4]

Normalization of lattices

Definition 8. Since we only care about Λ up to homothety, for a lattice $\langle p, q\rangle$ with $\tau = \frac{p}{q} \in \mathbb{H}$, we can normalize it to a homothetic lattice $\Lambda_\tau = \langle \tau, 1\rangle = \frac{1}{q}\langle p, q\rangle$. [3]

However, there might also be homothetic normalized lattices such as $\langle i, 1\rangle = \langle i+1, 1\rangle$. We identify them using the following theorem.

Theorem 9. Let $\tau, \tau' \in \mathbb{H}$. Λ_τ and $\Lambda_{\tau'}$ are homothetic if and only if τ and τ' are related with a Möbius map $\tau' = \frac{a\tau+b}{c\tau+d}$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}(2,\mathbb{Z})$.

Proof. By definition, Λ_τ and $\Lambda_{\tau'}$ are homothetic if and only if there is $z \in \mathbb{C} \setminus \{0\}$ such that $\Lambda_{\tau'} = z\Lambda_\tau$, which is equivalent to

$$\langle \tau', 1\rangle = \langle z\tau, z\rangle$$

which by Theorem 5 holds if and only if there is $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2,\mathbb{Z})$ such that

$$\begin{pmatrix} \tau' \\ 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} z\tau \\ z \end{pmatrix} \iff \tau' = \frac{a\tau+b}{c\tau+d}$$

□

This theorem points out the relationship between the action of $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$ and the classification of $\Lambda \subset \mathbb{C}$.

Specifically, if we use $\mathcal{L}$ to denote the set of all lattices in $\mathbb{C}$, and define an equivalence relation $\sim$ on L by $\Lambda \sim \Lambda'$ if and only if Λ and Λ' are homothetic, then there is a bijection from $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ to $\mathcal{L}/\sim$ given by

$$f : \tau \mapsto \Lambda_\tau$$

j-invariant

Definition 10. The j-invariant is a function from $\mathbb{H}$ to $\mathbb{C}$ defined by

$$j(\tau) = 1728\frac{g_2(\tau)^3}{\Delta(\tau)}$$

The core property of the function is that $j(\tau) = j(\tau')$ if and only if there is a Möbius map in $\mathrm{PSL}(2,\mathbb{Z})$ such that $\tau' = \frac{a\tau+b}{c\tau+d}$, or equivalently, Λ_τ is homothetic to $\Lambda_{\tau'}$ [3]. Therefore, j-invariant acts as an isomorphism from $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ to $\mathbb{C}$. On the other hand, when we apply the correspondence between $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ and $\mathcal{L}/\sim$ we just deduced into the j-invariant, we can, in fact, view it as a parameterization of each equivalence class to points of $\mathbb{C}$.

Summary

In this poster, we explored the action $\mathrm{PSL}(2,\mathbb{Z})$ on $\mathbb{H}$. We observed that the action of $\mathrm{PSL}(2,\mathbb{Z})$ tessellates $\mathbb{H}$ into copies of a fundamental domain, where any two triangles can be associated by a Möbius map in $\mathrm{PSL}(2,\mathbb{Z})$.
Then we studied the lattice structure in $\mathbb{C}$: any lattice is generated by independent $p, q \in \mathbb{C}$. While homothety provides a notion of similarity among lattices, it actually generates an equivalence relation, based on which one can classify all complex lattices.
We finally discovered that Λ_τ is homothetic to $\Lambda_{\tau'}$ if and only if τ' can be transformed from τ via a Möbius map in $\mathrm{PSL}(2,\mathbb{Z})$. It follows that the orbit space $\mathbb{H}/\mathrm{PSL}(2,\mathbb{Z})$ and the set of lattices up to homothety have the same structure and are connected by a bijection $\tau \mapsto \Lambda_\tau$. The j-invariant, as a parameterization of equivalence classes, gives us an algebraic view of this connection.

References

[1] J. S. Milne (2017) *Modular Functions and Modular Forms (v1.31)*, www.jmilne.org/math.

[2] Lozano-Robledo, Álvaro (2011) *Elliptic Curves, Modular Forms and their L-functions.* https://doi.org/10.1090/stml/058

[3] Silverman, J.H. (1994) *Advanced Topics in the Arithmetic of Elliptic Curves.* https://doi.org/10.1007/978-1-4612-0851-8

[4] Silverman, J. H. (2009). *The Arithmetic of Elliptic Curves.* https://doi.org/10.1007/978-0-387-09494-6